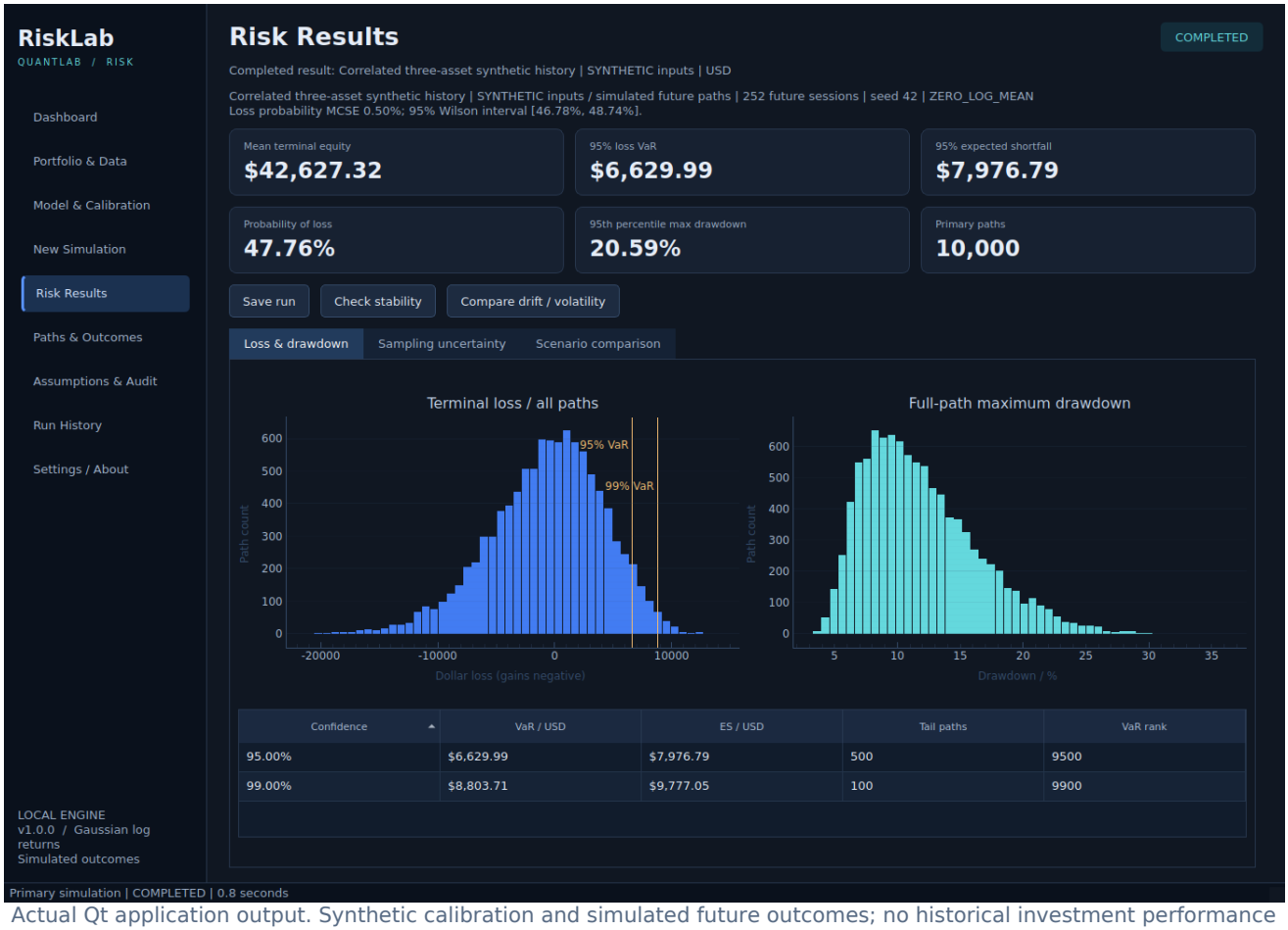
RiskLab

Engineering release report

QuantLab / Risk | Aiden Federico-Dyer | v1.0.0 | November 5, 2026

Complete source release of a local Monte Carlo portfolio risk desktop application. Verified numerical and desktop workflows, reproducible examples, auditable saved results and target-platform build scripts.

Passing tests	Headless baseline	Process peak RSS
102	0.59 seconds	32.0 MiB



Actual Qt application output. Synthetic calibration and simulated future outcomes; no historical investment performance is implied.

Platform status

Source and native packaged Linux offscreen workflows passed, including restart and saved-run reopening. Windows build/runtime/high-DPI checks remain UNVERIFIED. The source ZIP contains no Windows executable.

Model and risk definitions

The source design is unchanged. One to ten USD assets use fixed nonnegative fractional shares and zero-interest cash over at most 252 future sessions.

Contract	Implementation
Daily log returns	g ~ Normal(m, Sigma); S_next = S * exp(g). No extra half-variance subtraction.
Drift choice	Zero log mean by default, or estimated/declared daily log means. Zero log mean is not zero expected arithmetic return.
Calibration	Complete adjacent declared price sessions; sample covariance normalized by n - 1. No gaps or imputation.
Covariance	Scale-relative symmetry/PSD checks, recorded tiny corrections, singular PSD eigenfactor.
Valuation	Equity = fixed cash + sum(fixed quantity * simulated price), without display rounding.
Terminal risk	Loss = V0 - VN. Gains are negative. VaR/ES are not floored at zero.
Path risk	Maximum drawdown is measured against each path's own running peak, including step zero.

Exact acceptance fixtures

With V0 = $1,000 and terminal values $800, $900, $1,000, $1,100, $1,200, ascending losses are -$200, -$100, $0, $100, $200. At 80% confidence, VaR is $100, ES is $200 and probability of loss is 40%.

VaR uses one-based rank ceil(alpha*N); ES averages the largest ceil((1-alpha)*N) losses. Decimal confidence arithmetic gives exactly 100 tail paths at N = 10,000 and alpha = .99. Each selected production tail needs at least 100 paths.

The equity path 1000, 1200, 900, 1100 has maximum drawdown 25%, despite finishing with a 10% gain. These small fixtures validate definitions, not tail estimation accuracy.

Sampling uncertainty

Loss probability includes a plug-in Monte Carlo SE and 95% Wilson interval. Tail diagnostics use 300 whole-sample bootstrap resamples plus two independent repeats. The instability rule and independent stream derivations are exported. Paired drift/volatility comparisons are a separate experiment.

Verification and performance

The final suite has 102 passing tests, 0 failures, 0 errors and 0 skips. JUnit, raw benchmark and smoke reports accompany the release.

Requirement	Evidence
R1: valid data/covariance	Strict dates, units, schema, nonfinite-value, PSD and singular fixtures.
R2: asset order	Canonical symbol mappings and seeded permutation acceptance test.
R3: repeatability	Nonzero-volatility equal-seed array reproduction; preview RNG independence.
R4: loss vs drawdown	Hand fixtures and independent scalar reference at batch sizes 1, 257 and 512.
R5: assumptions/precision	Exported provenance, seeds, versions, tail counts, bootstrap intervals and repeats.
Statistical checks	One-asset lognormal moments and cross-asset means/covariance within declared sampling-SE tolerances.
Desktop/persistence	Real workers, event-loop responsiveness, cancel, sort/copy, corruption checks, portable reopen without fitting or sampling.

Measured host baseline

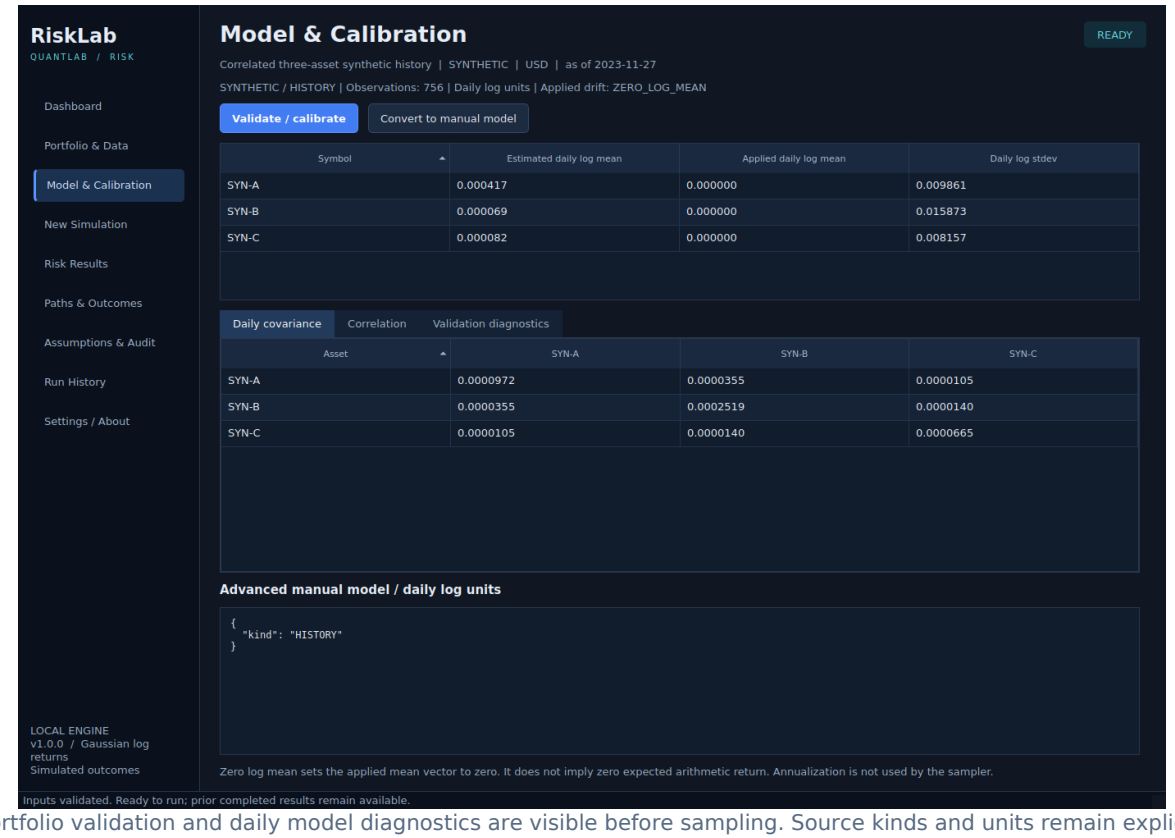
10,000 paths x 252 future steps x 10 assets; batch 512; 20 previews; seed 42. Wall time 0.589636 seconds; peak RSS 32.031 MiB. The 15-second / 300-decimal-MB headless design target was met on this host.

AMD EPYC 9V74 80-Core Processor, 9 visible CPUs, Linux; Python 3.12.14, NumPy 2.5.4, one OpenBLAS thread. Peak memory uses OS process RSS, not Python-only allocation tracing.

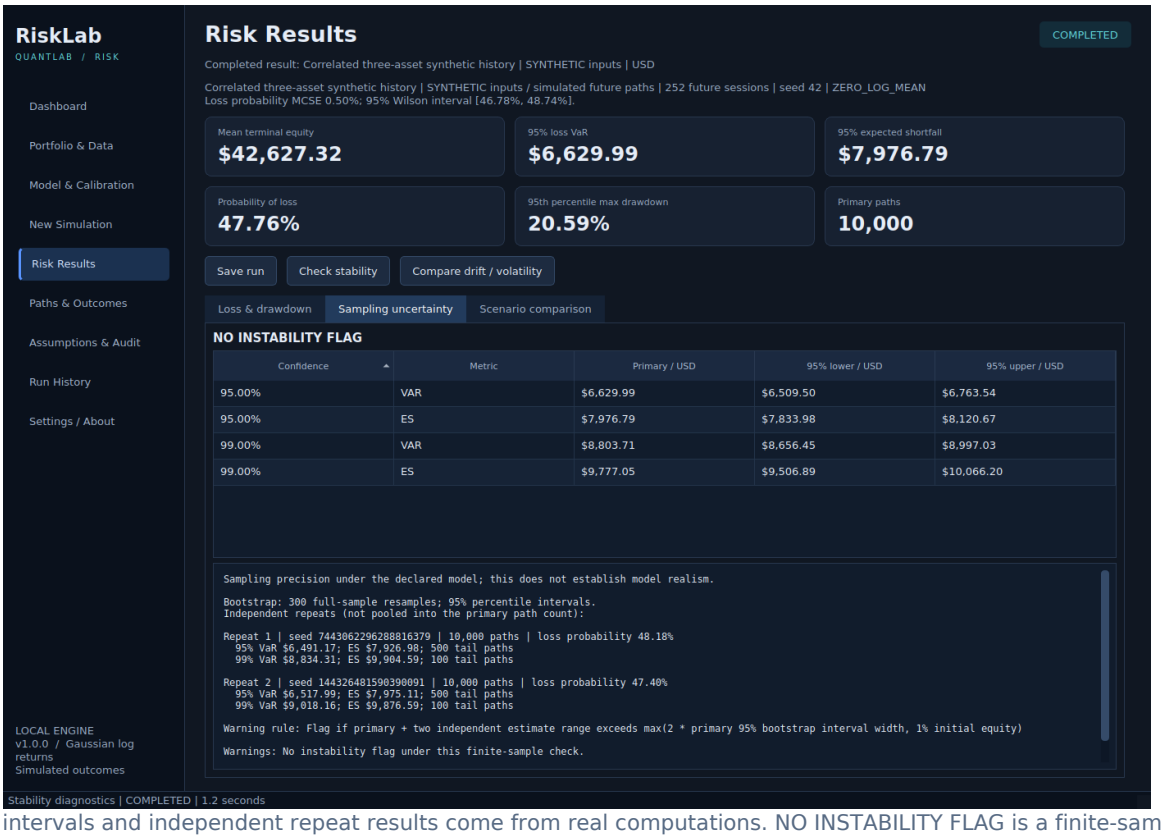
Phase	Wall time	Peak RSS
Source full Qt workflow	5.198 s	140.4 MiB
Packaged full Qt workflow	4.694 s	143.7 MiB

Full Qt checks use the three-asset synthetic demonstration and include diagnostics, comparisons, saving and screenshots. They differ from the ten-asset headless baseline. Native Windows and physical-display behavior remain unverified.

Integrated desktop evidence



Portfolio validation and daily model diagnostics are visible before sampling. Source kinds and units remain explicit.



Tail intervals and independent repeat results come from real computations. NO INSTABILITY FLAG is a finite-sample diagnostic outcome, not a guarantee.

Use, audit and limitations

Start on Windows

1. Extract the whole source ZIP and install Python 3.12 if needed.

2. Run Setup_Windows.cmd once, then Launch_RiskLab.cmd.

3. Load the synthetic demo, review its model and run the default simulation.

4. Check Stability, Compare drift / volatility, then Save Run.

Build_Windows.cmd runs tests, builds with PyInstaller and performs packaged smoke checks on Windows. Expected output: dist/RiskLab/RiskLab.exe. A normal native desktop/high-DPI check is still required.

Auditable saved results

Versioned folders store inputs, calibrated parameters, full outcomes, preview paths, summary, diagnostics, comparisons, sampling metadata and checksums. Writable-handle synchronization and verified atomic publication prevent failed/cancelled pre-commit saves becoming completed results.

Run History opens the included completed example or a relocated saved folder. Reopen uses stored outcomes and calibration, without fitting or stochastic execution. Diagnostics and comparisons create new in-memory revisions; saving creates a new folder, preserving earlier published runs.

Default synthetic example	Model output
Initial equity	$41,987.62
95% loss VaR / ES	$6,629.99 / $7,976.79
Loss probability / 95th-percentile max DD	47.76% / 20.59%

Limits and next platform check

Gaussian independent-session increments and constant covariance omit jumps, volatility clustering and changing dependence. No options, shorts, leverage, FX, rebalancing, execution, costs, dividends, splits, interest or automatic data downloads are modeled.

The next external verification step is a native Windows build and manual launch. Native Linux offscreen packaging passed. The source release includes test evidence, docs, real screenshots, completed examples, Windows scripts and a Git history bundle; no remote repository was created or pushed.

Technical sources: NumPy multivariate sampling, covariance and stream-compatibility documentation; Qt QThread documentation; PyInstaller operating-mode documentation. Full links, exact commands and detailed architecture are in the accompanying Markdown report.